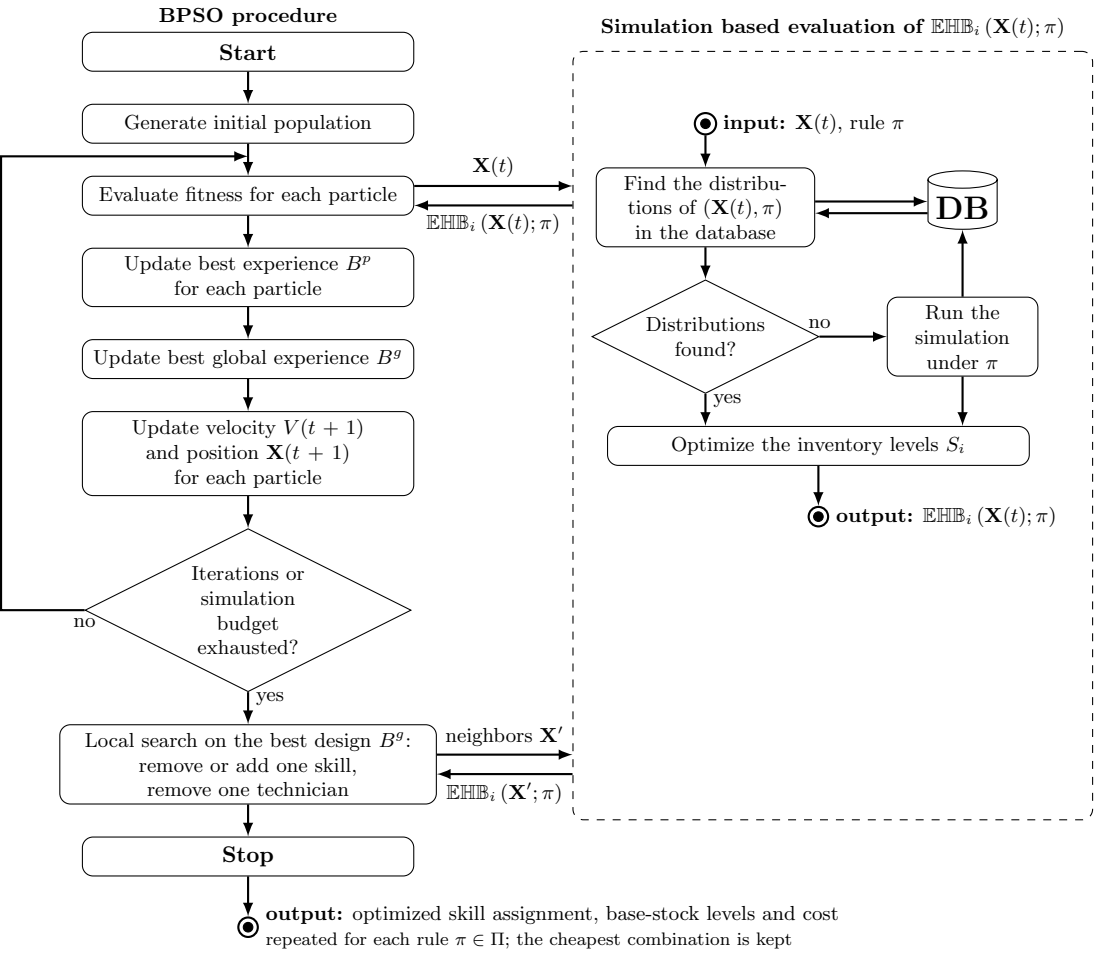

BPSO procedure
Start
Generate initial population
Evaluate fitness for each particle
Update best experience B^p for each particle
Update best global experience B^g
Update velocity $V(t+1)$ and position $\mathbf{X}(t+1)$ for each particle
Iterations or simulation budget exhausted?
no
yes
Local search on the best design B^g: remove or add one skill, remove one technician
Stop
output: optimized skill assignment, base-stock levels and cost
repeated for each rule $\pi \in \Pi$; the cheapest combination is kept
$\mathbf{X}(t)$
$\mathbb{EHB}_i(\mathbf{X}(t);\pi)$
neighbors $\mathbf{X}'$
$\mathbb{EHB}_i(\mathbf{X}';\pi)$
Simulation based evaluation of $\mathbb{EHB}_i(\mathbf{X}(t);\pi)$
input: $\mathbf{X}(t)$, rule π
Find the distributions of $(\mathbf{X}(t),\pi)$ in the database
DB
Distributions found?
no
yes
Run the simulation under π
Optimize the inventory levels S_i
output: $\mathbb{EHB}_i(\mathbf{X}(t);\pi)$